Statistical Computing

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Junyoung Byun

Spring 2026

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Statistical Computing

Lecturer

- **Junyoung Byun**
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- **Office:** 310-1048
- **Office hour:** By appointment
- **Research area:** AI Security & Privacy & Trust & Ethics

Scope

- Random Variate Generation
- Visualization of Multivariate Data
- Monte Carlo Integration
- Optimization Algorithms
- Constrained Optimization & Duality
- Parameter Estimation
- Kernels & Distances
- Density Estimation
- (Optional) Support Vector Machines

Lecture Format

- **Textbooks**
  - Maria L. Rizzo, *Statistical Computing with R, Second Edition*
  - Jaewook Lee, *AI 수학 교과서*
- **Lecture notes** (PDF) based on the textbooks
- Code practice will be based on **Python** (not R)

Prerequisites

- Mathematical Statistics
- Linear Algebra & Calculus
- Python programming (numpy, pandas, matplotlib + scipy, scikit-learn)

Assessment

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- **Assignments – 10%**
  - Approximately four assignments throughout the semester
  - May include short proofs, hand calculations, or coding exercises
  - **An F grade will be assigned** if two or more assignments are not submitted
  - Late submissions are accepted with a penalty

Assessment

- **Midterm – 45%** Week 8
- **Final – 45%** Week 16
- No attendance score (up to 4 absences)
- Attendance may be checked randomly (5–10 times during the semester)
- **※ An F grade will be assigned for more than 4 absences**
- **Electronic attendance will not be accepted**